

Module 3(a) - Properties of DFT and its Applications

DFT Properties: Time reversal of a sequence, Circular Time shift of a sequence, Circular frequency shift, Complex conjugate property, Multiplication of two sequences, Parseval's theorem. Linear Filtering methods based on the DFT.

Additional DFT Properties

Time Reversal of a Sequence

Reversing the N point sequence in time is equivalent to reversing the DFT sequence

If $x(n) \xleftarrow{\text{DFT}} X(k)$, then $x((-n))_N = x(N - n) \xleftarrow{\text{DFT}} X((-k))_N = X(N - k)$

Proof: $\text{DFT}[x(n)] = \sum_{n=0}^{N-1} x(n) e^{-j\left(\frac{2\pi kn}{N}\right)}$

$$\text{DFT}[x(N - n)] = \sum_{n=0}^{N-1} x(N - n) e^{-j\left(\frac{2\pi k(N-n)}{N}\right)} = \sum_{n=0}^{N-1} x(N - n) e^{j\left(\frac{2\pi kn}{N}\right)} e^{-j2\pi k} = \sum_{n=0}^{N-1} x(N - n) e^{j\left(\frac{2\pi kn}{N}\right)}$$

Let $m = N - n$ so that $n = N - m$, When $n = 0$; $m = N$ and when $n = N - 1$; $m = N - (N - 1) = 1$

Using these values in the above equation

$$\text{DFT}[x(N - n)] = \sum_{m=N}^1 x(m) e^{j\frac{2\pi km}{N}} (1) = \sum_{m=1}^N x(m) e^{j\frac{2\pi km}{N}} \left(e^{-j\frac{2\pi mN}{N}} \right) = \sum_{m=0}^{N-1} x(m) e^{-j\frac{2\pi m(N-k)}{N}} = X(N - k)$$

Note: $X^*(N - k) = \sum_{m=0}^{N-1} [x(m) e^{-j2\pi k(N-m)/N}]^* = \sum_{m=0}^{N-1} x^*(m) e^{j2\pi km/N}$

Circular Time Shift

If the discrete time signal is circularly shifted in time by **m** units, then its DFT is multiplied by $e^{-j2\pi km/N}$

If $x(n) \xleftarrow{\text{DFT}} X(k)$ then $x((n - m))_N \xleftarrow{\text{DFT}} X(k) e^{-j2\pi km/N}$

Proof: $\text{DFT}[x(n)] = \sum_{n=0}^{N-1} x(n) e^{-j\left(\frac{2\pi kn}{N}\right)}$

$$\text{DFT}[x((n - m))_N] = \sum_{n=0}^{N-1} x((n - m))_N e^{-j\left(\frac{2\pi kn}{N}\right)}$$

Let $p = (n - m)$ & $n = (p + m)$. When $n = 0$; $p = -m$ and when $n = N - 1$; $p = (-m + N - 1)$

Using these values in the above equation

$$\text{DFT}[x((n - m))_N] = \sum_{p=-m}^{-m+N-1} x(p) e^{-j\frac{2\pi k(p+m)}{N}} = \left[\sum_{p=0}^{N-1} x(p) e^{-j\frac{2\pi kp}{N}} \right] e^{-j\frac{2\pi km}{N}} = X(k) e^{-j\frac{2\pi km}{N}} = X(k) W_N^{km}$$

Circular Frequency Shift :

If a discrete time signal is multiplied by $e^{j2\pi mn/N}$, its DFT is circularly shifted by m units

If $x(n) \xleftrightarrow{\text{DFT}} X(k)$ then $x(n) e^{j2\pi mn/N} \xleftrightarrow{\text{DFT}} X(k-m)_N$

Proof: $\text{DFT}[x(n)] = \sum_{n=0}^{N-1} x(n) e^{-j\left(\frac{2\pi kn}{N}\right)}$

$$\text{DFT}\left[x(n)e^{j\frac{2\pi mn}{N}}\right] = \sum_{n=0}^{N-1} x(n) e^{-j\left(\frac{2\pi kn}{N}\right)} e^{j\frac{2\pi mn}{N}} = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}(k-m)n} = X((k-m))_N$$

Multiplication of Two Sequences (Modulation Property)

The DFT of the product of two discrete time sequences is equivalent to circular convolution of the DFTs of the individual sequences scaled by a factor $(1/N)$

If $x(n) \xleftrightarrow{\text{DFT}} X(k)$, then $[x_1(n).x_2(n)] = \frac{1}{N}[X_1(k) \odot X_2(k)]$

Proof: By definition of DFT

$$x_1(n) = \frac{1}{N} \sum_{k=0}^{N-1} X_1(k) e^{j\left(\frac{2\pi kn}{N}\right)} = \frac{1}{N} \sum_{m=0}^{N-1} X_1(m) e^{j\left(\frac{2\pi mn}{N}\right)} \quad \text{--- (put } k = m \text{)}$$

$$\text{DFT}[x_1(n)x_2(n)] = \sum_{n=0}^{N-1} [x_1(n)x_2(n)] e^{-j\left(\frac{2\pi kn}{N}\right)} = \sum_{n=0}^{N-1} \left[\frac{1}{N} \sum_{m=0}^{N-1} X_1(m) e^{j\left(\frac{2\pi mn}{N}\right)} \right] x_2(n) e^{-j\left(\frac{2\pi kn}{N}\right)}$$

Changing the order of summation

$$= \frac{1}{N} \sum_{m=0}^{N-1} X_1(m) \left[\sum_{n=0}^{N-1} x_2(n) e^{-j\left(\frac{2\pi kn}{N}\right)} e^{j\left(\frac{2\pi mn}{N}\right)} \right] = \frac{1}{N} \sum_{m=0}^{N-1} X_1(m) \left[\sum_{n=0}^{N-1} x_2(n) e^{-j\frac{2\pi(k-m)n}{N}} \right]$$

$$= \frac{1}{N} \left[\sum_{m=0}^{N-1} X_1(m) X_2((k-m))_N \right] \quad \text{--- (I)}$$

Equation (I) represents the circular convolution, hence $[x_1(n).x_2(n)] = \frac{1}{N}[X_1(k) \odot X_2(k)]$

➔ Multiplication in time domain is equivalent to Convolution in frequency domain ➔

1. Find $Y(k)$ if $x_1(n) = 1; 0 \leq n \leq 7$, $x_2(n) = \cos(\pi n/4); 0 \leq n \leq 7$ and $y(n) = x_1(n).x_2(n)$

Sol – $x(n) = [1, 1, 1, 1, 1, 1, 1, 1]$

$$X_1(k) = \sum_{n=0}^{N-1} x_1(n) W_N^{kn} = \sum_{n=0}^7 [1 W_8^k]^n = \frac{W_8^{8k} - 1}{W_8^k - 1} = \begin{cases} 0; & k \neq 0 \\ 8; & k = 0 \end{cases}$$

$$X_1(k) = \{8, 0, 0, 0, 0, 0, 0, 0\}$$

$$x_2(n) = \cos\left(\frac{n\pi}{4}\right) = \left[\frac{e^{j\frac{n\pi}{4}} + e^{-j\frac{n\pi}{4}}}{2} \right] = \left[\frac{e^{j\frac{2\pi n}{8}} + e^{-j\frac{2\pi n}{8}}}{2} \right] = \left[\frac{W_8^{-kn} + W_8^{kn}}{2} \right]$$

$$X_2(k) = N \left[\frac{\delta(k+1) + \delta(k-1)}{2} \right] = 8 \left[\frac{\delta(k+1) + \delta(k-1)}{2} \right]$$

$$X_2(k) = 4[\delta(k+1) + \delta(k-1)] = 4[\delta(k-1) + \delta(k-7)]$$

$\delta(k+1)$ has value at $k = -1$
for 8 point DFT, $k = -1$
is same as $k = 8 - 1 = 7$,
hence $\delta(k+1) = \delta(k-7)$

Using multiplication property of DFT

$$y(n) = x_1(n) \cdot x_2(n) \xrightarrow{\text{DFT}} Y(k) = \frac{1}{N} (X_1(k) \otimes_N X_2(k))$$

$$Y(k) = \frac{1}{8} [X_1(k) \otimes_N X_2(k)] = \frac{1}{8} [X_1(k) \otimes_N 4[\delta(k-1) + \delta(k-7)]] = \frac{4}{8} [X_1(k-1) + X_1(k-7)]$$

$$Y(k) = \frac{1}{2} [0, 8, 0, 0, 0, 0, 0, 8] = \{0, 4, 0, 0, 0, 0, 0, 4\}$$

2. If $x(n) = \{7, 0, 8, 0\}$, find $y(n)$ if $Y[k] = X((k-2))_4$

Solution: $Y[k] = X((k-2))_4$ Given $N = 4$; $m = 2$

From circular frequency shift property $x(n)e^{j2\pi nm/N} \xrightarrow{-\text{DFT}} X(k-m)$

$$y(n) = \text{IDFT}[X((k-2))_4] = e^{\frac{j2\pi 2n}{4}} x(n) = x(n)e^{j2\pi 2n/4} = x(n)e^{j\pi n} = (-1)^n x(n) = \{7, 0, 8, 0\}$$

3. If $g(n)$ and $h(n)$ are the two sequences of length 6 with 6 point DFT's $G[k]$ and $H[k]$ respectively. Given $g(n) = \{4, 3, 1, 5, 2, 6\}$ and $H[k] = G[[k-3]]_6$. Determine $h(n)$ without computing DFT and IDFT

Solution: $H[k] = G[[k-3]]_6$ Given $N = 6$; $m = 3$

From Circular frequency shift property $x(n)e^{j2\pi nm/N} \xrightarrow{-\text{DFT}} X((k-m))_N$

$$h(n) = \text{IDFT}[G((k-3))_6] = e^{\frac{j2\pi 3n}{6}} g(n) = g(n)e^{j2\pi 3n/6} = g(n)e^{j\pi n} = (-1)^n x(n) = \{4, -3, 1, -5, 2, -6\}$$

4. Two finite sequences $x(n) = \{x(0), x(1), x(2), x(3)\}$ and $h(n) = \{h(0), h(1), h(2), h(3)\}$ have DFT's given by $X[k] = \{1, j, -1, -j\}$; $H[k] = \{0, 1+j, 1, 1-j\}$. Use the properties of the DFT and find the following: (i) $X_1[k] = \text{DFT}\{h(0), -h(1), h(2), -h(3)\}$ (ii) DFT of $y(n) = x(n) \otimes_4 h(n)$ (iii) $X_3[k] = \text{DFT}\{x(0), h(0), x(1), h(1), x(2), h(2), x(3), h(3)\}$

Solution: (i) $X[k] = \{1, j, -1, -j\}$ $H[k] = \{0, (1+j), 1, (1-j)\}$

$$x_1(n) \xrightarrow{\text{DFT}_N} X_1[k] = \text{DFT}\{h(0), -h(1), h(2), -h(3)\}$$

$$X_1[k] = \text{DFT}[(-1)^n h(n)] = \text{DFT}\{e^{j\pi n} h(n)\} = \text{DFT}[e^{\frac{j2\pi 2n}{4}} h(n)]$$

Using circular frequency shift property

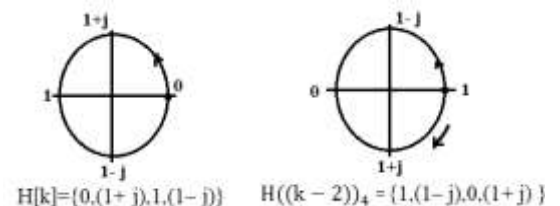
$$x(n)e^{j2\pi nm/N} \xrightarrow{-\text{DFT}} X((k-m))_N \quad \text{Here } m = 2, N = 4$$

$$X_1[k] = H((k-2))_4 = \{1, (1-j), 0, (1+j)\}$$

(ii) DFT of $y(n) = x(n) \otimes_4 h(n)$

From circular convolution property

$$Y(k) = X[k] \cdot H[k] = \{1, j, -1, -j\} \cdot \{0, 1+j, 1, 1-j\} = \{0, -1+j, -1-j\}$$



$$(iii) X_3[k] = \text{DFT}\{x(0), h(0), x(1), h(1), x(2), h(2), x(3), h(3)\}$$

$$X_3[k] = \text{DFT}[x_3(n)] = \text{DFT}\{x_3(0), x_3(1), x_3(2), x_3(3), x_3(4), x_3(5), x_3(6), x_3(7)\}$$

Divide the summation into even and odd components

$$X_3[k] = \sum_{n=0}^{N-1} x_3(n) e^{-j2\pi kn/N} = \sum_{n=0}^7 x_3(n) W_8^{kn} = \sum_{m=0}^3 x_3(2m) W_8^{2km} + \sum_{m=0}^3 x_3(2m+1) W_8^{k(2m+1)}$$

$$X_3[k] = \sum_{n=0}^3 x(n) [W_8^2]^{kn} + W_8^k \sum_{n=0}^3 h(n) [W_8^2]^{kn} = \sum_{n=0}^3 x(n) W_4^{kn} + W_8^k \sum_{n=0}^3 h(n) W_4^{kn}$$

$$X_3[k] = X(k) + W_8^k H(k) \quad k = 0, 1, 2, 3$$

$$X_3[0] = X[0] + H[0] = 1;$$

$$X_3[1] = X[1] + W_8^1 H[1] = j + (0.707 - j 0.707)(1 + j) = 1.414 + j$$

$$X_3[2] = X[2] + W_8^2 H[2] = -1 + (-j)(1) = -1 - j$$

$$X_3[3] = X[3] + W_8^3 H[3] = -j + (-0.707 - j 0.707)(1 - j) = -1.414 - j$$

$$X_3[4] = X[4] + W_8^4 H[4] = X(0) + W_8^4 H(0) = 1 - 1(0) = 1$$

$$X_3[5] = X[1] + W_8^1 H[1] = j - (0.707 - j 0.707)(1 + j) = -1.414 + j$$

$$X_3[6] = X[2] + W_8^2 H[2] = -1 - (-j)(1) = -1 + j$$

$$X_3[7] = X[3] + W_8^3 H[3] = -j - (-0.707 - j 0.707)(1 - j) = 1.414 - j$$

5. If $h(n) = \begin{cases} \frac{1}{2}; & -2 \leq n \leq 2 \\ 0; & \text{otherwise} \end{cases}$ Determine the 8 point DFT of $h(n)$.

Solution: $h(n) = \left\{ \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2} \right\} = \left\{ \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0 \right\}$

Let $h_p(n) = h(n+2)$ shift $h(n)$ 2 samples in the clockwise direction

$$h_p(n) = \left\{ \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, \frac{1}{2}, \frac{1}{2} \right\}$$

$$H_p(k) = \sum_{n=0}^{N-1} h_p(n) W_N^{kn} = \sum_{n=0}^7 h_p(n) W_8^{kn} = \frac{1}{2} [1 + W_8^k + W_8^{2k} + W_8^{6k} + W_8^{7k}]$$

$$k=0; H_p(0) = \frac{1}{2} [1 + 1 + 1 + 1 + 1] = 2.5$$

$$k=1; H_p(1) = \frac{1}{2} [1 + W_8^1 + W_8^2 + W_8^6 + W_8^7] = \frac{1}{2} [1 + (0.707 - j0.707) - j + j + (0.707 + j0.707)] = 1.207$$

$$k=2; H_p(2) = \frac{1}{2} [1 + W_8^2 + W_8^4 + W_8^{12} + W_8^{14}] = \frac{1}{2} [1 - j - 1 - 1 + j] = -0.5$$

$$k=3; H_p(3) = \frac{1}{2} [1 + W_8^3 + W_8^6 + W_8^{18} + W_8^{21}] = \frac{1}{2} [1 + (-0.707 - j0.707) + j - j + (-0.707 + j0.707)] = -0.207$$

The spectrum of $h(n)$ is periodic and continuous. If we select the data points such that the IDFT of the spectrum gives $h_p(n)$ such that

$$h_p(n) = \left\{ \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, \frac{1}{2}, \frac{1}{2} \right\}$$

$$k=4; H_p(4) = \frac{1}{2} [1 + W_8^4 + W_8^8 + W_8^{24} + W_8^{28}] = \frac{1}{2} [1 + (-1) + 1 + 1 + (-1)] = 0.5$$

From symmetry property of real valued sequence $X(k) = X^*(N - k)$ here $N = 8$

$$H_p(5) = H_p^*(8 - 5) = H_p^*(3) = -0.207 \quad H_p(6) = H_p^*(8 - 6) = H_p^*(2) = -0.5 \quad \text{and } H_p(7) = H_p^*(1) = 1.207$$

$$H(k) = \{ 2.5, 1.207, -0.5, -0.207, 0.5, -0.207, -0.5, 1.207 \}$$

Using circular time shift property $x((n + m))_N \xleftarrow{\text{DFT}} e^{-j2\pi km/N} X(k)$

$$\text{Here } N = 8; m = 2 \quad H(k) = e^{-\frac{2\pi km}{N}} H_p(k) = e^{-\frac{2\pi k \cdot 2}{8}} H(k) = e^{-\frac{\pi k}{2}} H(k); \text{ --- } (k = 0 \text{ to } 7)$$

$$X(k) = \{ 2.5, j1.207, 0.5, -j0.207, 0.5, -j0.207, -0.5, j1.207 \}$$

6. If $x(n) = 4\delta(n) + 3\delta(n - 1) + 2\delta(n - 2) + \delta(n - 3)$; Compute the 6 point DFT of the signal $x(n)$ and also determine finite length sequence $y(n)$ such that $Y(k) = W_6^{4k} X(k)$

Solution: $x(n) = \{ 4, 3, 2, 1 \}$ $N = 6$; hence $x(n) = \{ 4, 3, 2, 1, 0, 0 \}$

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} = \sum_{n=0}^5 x(n) W_6^{kn} = 4 + 3W_6^k + 2W_6^{2k} + W_6^{3k}$$

$$X(0) = 4 + 3W_6^0 + 2W_6^{2(0)} + W_6^{3(0)} = 10$$

$$X(1) = 4 + 3W_6^1 + 2W_6^2 + W_6^3 = 3.5 - j4.33$$

$$X(2) = 4 + 3W_6^2 + 2W_6^{2(2)} + W_6^{3(2)} = 2.5 - j0.866$$

$$X(3) = 4 + 3W_6^3 + 2W_6^{2(3)} + W_6^{3(3)} = 2$$

$$X(4) = 4 + 3W_6^4 + 2W_6^{2(4)} + W_6^{3(4)} = 2.5 + j0.866$$

$$X(5) = 4 + 3W_6^5 + 2W_6^{2(5)} + W_6^{3(5)} = 3.5 + j4.33$$

$$X(k) = \{ 10, (3.5 - j4.33), (2.5 - j0.866), -2, (2.5 + j0.866), (3.5 + j4.33) \}$$

$$Y(k) = W_6^{4k} X(k) = e^{-\frac{j2\pi k(4)}{6}} X(k)$$

Taking IDFT on both sides and using time shifting property $x((n - m))_N \xleftarrow{\text{DFT}} e^{-\frac{j2\pi km}{N}} X(k)$

here $m = 4; N = 6 \quad y(n) = x((n - 4))_6 = \{ 2, 1, 0, 0, 4, 3 \}$ (rotate $x(n)$ by 4 samples in ACW direction)

$$\begin{aligned} W_6^0 &= 1 & W_6^1 &= e^{-\frac{j2\pi}{6}} = 0.5 - j0.866 \\ W_6^2 &= e^{-\frac{j2\pi(2)}{6}} = (0.5 - j0.866) = W_6^8 \\ W_6^3 &= e^{-\frac{j2\pi(3)}{6}} = -1 = W_6^9 = W_6^{15} \\ W_6^4 &= e^{-\frac{j2\pi(4)}{6}} = (-0.5 - j0.866) \\ W_6^5 &= e^{-\frac{j2\pi(5)}{6}} = (0.5 + j0.866) \end{aligned}$$

7. Using time shifting property, find the 8 point DFT of the sequence $x(n) = \delta(n) + \delta(n - 4) + 2\delta(n - 5)$

Solution: $\delta(n) \xleftarrow{\text{DFT}} 1$ Using time shifting property $x((n - m))_N \xleftarrow{\text{DFT}} e^{-\frac{j2\pi km}{N}} X(k)$

$$x(n) = \delta(n) + \delta(n - 4) + 2\delta(n - 5) = [1, 0, 0, 0, 1, 2, 0, 0]$$

here $N = 8$; also for second term $m = 4$ and for third term $m = 5$

$$X(k) = 1 + e^{-\frac{j2\pi(4)k}{8}} (1) + 2e^{-\frac{j2\pi(5)k}{8}} (1) = 1 + W_8^{4k} + 2W_8^{5k} \text{ ----- } k = 0 \text{ to } 7$$

$$X(0) = 1 + W_8^0 + 2W_8^0 = 4$$

$$X(1) = 1 + W_8^4 + 2W_8^5 = 1 + (-1) + 2(-0.707 + j0.707) = (-1.414 + j1.414)$$

$$X(2) = 1 + W_8^8 + 2W_8^{10} = 1 + W_8^0 + 2W_8^2 = 1 + (1) + 2(-j) = 2 - j2$$

$$X(3) = 1 + W_8^{12} + 2W_8^{15} = 1 + W_8^4 + 2W_8^7 = 1 + (-1) + 2(0.707 + j0.707) = (1.414 + j1.414)$$

$$X(4) = 1 + W_8^{16} + 2W_8^{20} = 1 + W_8^0 + 2W_8^4 = 1 + (1) + 2(-1) = 0$$

$$X(5) = 1 + W_8^{20} + 2W_8^{25} = 1 + W_8^4 + 2W_8^1 = 1 + (-1) + 2(0.707 - j0.707) = 1.414 - j1.414$$

$$X(6) = 1 + W_8^{24} + 2W_8^{30} = 1 + W_8^0 + 2W_8^6 = 1 + (1) + 2(j) = 2 + j2$$

$$X(7) = 1 + W_8^{28} + 2W_8^{35} = 1 + W_8^4 + 2W_8^3 = 1 + (-1) + 2(-0.707 - j0.707) = (-1.414 - j1.414)$$

$$X(k) = \{ 10, (-1.414 + j1.414), (2 - j2), (1.414 + j1.414), 0, (1.414 - j1.414), (2 + j2), (-1.414 - j1.414) \}$$

8. If $X(k) = \{ 10, (-2 + j2), -2, (-2 - j2) \}$ Using the properties of DFT, find the DFT's of the following sequences (i) $x_1(n) = x((n-2))_4$ (ii) $x_2(n) = x((4-n))_4$

Solution: (i) $x_1(n) = x((n-2))_4$ Using circular time shift property, here $N = 4, m = 2$

$$X_1(k) = e^{\frac{-j2\pi km}{N}} X(k) = e^{\frac{-j2\pi k2}{4}} X(k) = (e^{-j\pi k})X(k) = (-1)^k X(k) = \{10, (2 - j2), 2, (2 + j2)\}$$

(ii) $x_2(n) = x((4-n))_4$ Using circular time reversing property,

$$x((-n))_N = x((N-n))_N \xleftarrow{\text{DFT}} X(N-k) \quad \text{here } N = 4,$$

$$X_2(k) = X(4-k) = \{10, (-2 - j2), -2, (-2 + j2)\}$$

9. Find the 4 point DFT of $x(n) = \sin^2\left(\frac{n\pi}{2} + \frac{\pi}{3}\right)$ & from $X(k)$, obtain the DFT of $y(n) = \cos^2\left(\frac{n\pi}{2} + \frac{\pi}{3}\right)$

Solution: $x(n) = \sin^2\left(\frac{n\pi}{2} + \frac{\pi}{3}\right) = \{ 0.75, 0.25, 0.75, 0.25 \}$

$$y(n) = \cos^2\left(\frac{n\pi}{2} + \frac{\pi}{3}\right) = \{ 0.25, 0.75, 0.25, 0.75 \}$$

Comparing $x(n)$ and $y(n)$, it can be concluded that $y(n) = x((n-1))_4 = x((n-1))_4$

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} = \sum_{n=0}^3 x(n) W_4^{kn} = 0.75 + 0.25W_4^k + 0.75W_4^{2k} + 0.25W_4^{3k} = \{2, 0, 1, 0\}$$

$$y(n) = x((n-1))_4 = x((n-1))_4$$

Taking DFT on both sides and using circular time shifting property $x((n-m))_N \xleftarrow{\text{DFT}} e^{\frac{-j2\pi km}{N}} X(k)$

$$\text{Here } N = 4; m = 1 \quad Y(k) = e^{\frac{-j2\pi k1}{4}} X(k) = e^{\frac{-j\pi k}{2}} X(k) = \{2, 0, -1, 0\}$$

10. If $X(k) = \{ 0, (1+j), 1, (1-j) \}$ Find the DFT of the following signals

(i) $x_1(n) = e^{\frac{-jn\pi}{2}} x(n)$ (ii) $x_2(n) = \cos\left(\frac{n\pi}{2}\right) x(n)$ (iii) $x_3(n) = x((n-1))_4$

Solution : $x_1(n) = e^{\frac{-jn\pi}{2}} x(n) = e^{\frac{-j2\pi(1)n}{4}} x(n)$

Using circular frequency shift property $e^{\frac{j2\pi mn}{N}} x(n) \xleftarrow{\text{DFT}} X((k-m))_N$ here $m = 1$; $N = 4$;

Hence $X_1(k) = X(k-1)_4 = \{ (1-j), 0, (1+j), 1 \}$

(ii) $x_2(n) = \cos\left(\frac{n\pi}{2}\right) x(n) = \left(\frac{e^{\frac{jn\pi}{2}} + e^{\frac{-jn\pi}{2}}}{2}\right) x(n) = \frac{1}{2} \left(e^{\frac{j2\pi n}{4}} x(n) + e^{\frac{-j2\pi n}{4}} x(n) \right)$

Using circular frequency shift property $e^{\frac{j2\pi mn}{N}} x(n) \xleftarrow{\text{DFT}} X((k-m))_N$

For first term $m = 1$; $N = 4$; and for second term $m = -1$

$X_2(k) = \frac{1}{2} (X(k-1) + X(k+1)) = \frac{1}{2} [\{ (1-j), 0, (1+j), 1 \} + \{ (1+j), 0, (1-j), 1 \}] = \left\{ 1, \frac{1}{2}, 1, \frac{1}{2} \right\}$

(iii) $x_3(n) = x((n-1))_4$ here $m = 1$; $N = 4$

Using circular time shifting property $x((n-m))_N \xleftarrow{\text{DFT}} e^{-\frac{j2\pi km}{N}} X(k)$

$X_3(k) = e^{-\frac{j2\pi k1}{4}} X(k) = e^{-\frac{j2\pi k}{4}} X(k)$

$X_3(0) = e^{-\frac{j2\pi(0)}{4}} X(0) = X(0) = 0$

$X_3(1) = e^{-\frac{j2\pi(1)}{4}} X(1) = -jX(1) = (1-j)$

$X_3(2) = e^{-\frac{j2\pi(2)}{4}} X(2) = X(2) = -1$

$X_3(3) = e^{-\frac{j2\pi(3)}{4}} X(3) = jX(3) = (1+j)$

$X_3(k) = \{ 0, (1-j), -1, (1+j) \}$

11. If $X(k)$ is the DFT of the sequence $x(n)$; determine the N point DFT of the sequence

(i) $x_1(n) = x(n) \cos\left(\frac{2\pi mn}{N}\right)$; $0 \leq n \leq (N-1)$ (ii) $x_2(n) = x(n) \sin\left(\frac{2\pi mn}{N}\right)$; $0 \leq n \leq (N-1)$

Solution: (i) $x_1(n) = x(n) \cos\left(\frac{2\pi mn}{N}\right) = x(n) \left[\frac{e^{j2\pi mn} + e^{-j2\pi mn}}{2} \right]$

$X_1(k) = \sum_{n=0}^{N-1} x_1(n) e^{-\frac{j2\pi kn}{N}} = \sum_{n=0}^{N-1} x(n) \left[\frac{e^{j2\pi mn} + e^{-j2\pi mn}}{2} \right] e^{-\frac{j2\pi kn}{N}}$

$X_1(k) = \frac{1}{2} \sum_{n=0}^{N-1} \left(x(n) e^{\frac{j2\pi(m)n}{N}} \right) e^{-\frac{j2\pi kn}{N}} + \frac{1}{2} \sum_{n=0}^{N-1} \left(x(n) e^{\frac{j2\pi(-m)n}{N}} \right) e^{-\frac{j2\pi kn}{N}}$

Using circular frequency shift property $e^{\frac{j2\pi mn}{N}} x(n) \xleftarrow{\text{DFT}} X((k-m))_N$

$X_1(k) = \left(\frac{X(k-m)_N + X(k+m)_N}{2} \right)$

(ii) $x_2(n) = x(n) \sin\left(\frac{2\pi mn}{N}\right)$

$X_2(k) = \sum_{n=0}^{N-1} x_2(n) e^{-\frac{j2\pi kn}{N}} = \sum_{n=0}^{N-1} x(n) \left[\frac{e^{j2\pi mn} - e^{-j2\pi mn}}{2j} \right] e^{-\frac{j2\pi kn}{N}}$

$$X_2(k) = \frac{1}{2j} \sum_{n=0}^{N-1} \left(x(n) e^{j\frac{2\pi(m)n}{N}} \right) e^{-j\frac{2\pi kn}{N}} - \frac{1}{2j} \sum_{n=0}^{N-1} \left(x(n) e^{j\frac{2\pi(-m)n}{N}} \right) e^{-j\frac{2\pi kn}{N}}$$

Using circular frequency shift property $e^{j\frac{2\pi mn}{N}} x(n) \xleftarrow{\text{DFT}} \rightarrow X((k-m))_N$

$$X_2(k) = \left(\frac{X((k-m))_N - X((k+m))_N}{2j} \right)$$

12. Given the sequence $x(n) = \begin{bmatrix} 1; & 0 \leq n \leq 3 \\ 0; & 4 \leq n \leq 7 \end{bmatrix}$ Compute the DFT of $x_1(n) = \begin{bmatrix} 1; & n = 0 \\ 0; & 1 \leq n \leq 4 \\ 1; & 5 \leq n \leq 7 \end{bmatrix}$

Solution: Given $x(n) = \{ 1, 1, 1, 1, 0, 0, 0, 0 \}$ $x_1(n) = \{ 1, 0, 0, 0, 0, 1, 1, 1 \}$

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} = \sum_{n=0}^7 x(n) W_8^{kn} = 1 + W_8^k + W_8^{2k} + W_8^{3k} \quad \text{--- (0} \leq k \leq 7)$$

$$X(k) = \{ 4, (1 - j2.414), 0, (1 - j0.414), 0, (1 + j0.414), 0, (1 + j2.414) \}$$

Comparing the sequences, $x_1(n) = x((n+3))_8$

Taking DFT on both sides and using time shifting property $x((n-m))_N \xleftarrow{\text{DFT}} \rightarrow e^{-j\frac{2\pi km}{N}} X(k)$

$$\text{Here } m = -3; N = 8 \quad X_1(k) = e^{-j\frac{2\pi k(3)}{8}} X(k) = e^{-j\frac{3\pi k}{4}} X(k) \quad \text{--- (0} \leq k \leq 7)$$

$$X_1(k) = \{ 4, (1 + j2.414), 0, (1 + j0.414), 0, (1 - j0.414), 0, (1 - j2.414) \}$$

Complex Conjugate Property

(a) If $x(n) \xleftarrow{\text{DFT}} \rightarrow X(k)$

Then $x^*(n) \xleftarrow{\text{DFT}} \rightarrow X^*((-k))_N = X^*(N-k)$

Proof: By definition DFT of $x^*(n)$ is given as

$$\text{DFT}[x^*(n)] = \sum_{n=0}^{N-1} x^*(n) e^{-j\left(\frac{2\pi kn}{N}\right)} = \sum_{n=0}^{N-1} x^*(n) e^{-j\left(\frac{2\pi kn}{N}\right)} e^{j\left(\frac{2\pi nN}{N}\right)} \quad \text{--- (since } e^{j2\pi n} = 1 \text{ always)}$$

$$\text{DFT}[x^*(n)] = \sum_{n=0}^{N-1} x^*(n) e^{j\frac{2\pi n(N-k)}{N}} = \left[\sum_{n=0}^{N-1} [x(n) e^{-j\frac{2\pi n(N-k)}{N}}] \right]^* = [X(N-k)]^* = X^*[N-k]$$

(b) If $x(n) \xleftarrow{\text{DFT}} \rightarrow X(k)$

Then $x^*(-n) = x^*(N-n) \xleftarrow{\text{DFT}} \rightarrow X^*(k)$

Proof:

$$\text{IDFT}[X^*(K)] = \frac{1}{N} \sum_{k=0}^{N-1} X^*(k) e^{j\left(\frac{2\pi kn}{N}\right)}$$

$$\text{IDFT}[X^*(K)] = \frac{1}{N} \sum_{k=0}^{N-1} \left[X(k) e^{-j\left(\frac{2\pi kn}{N}\right)} \right]^* = \frac{1}{N} \sum_{k=0}^{N-1} \left[X(k) e^{-j\left(\frac{2\pi kn}{N}\right)} e^{j\left(\frac{2\pi kN}{N}\right)} \right]^* \quad - (\text{since } e^{j2\pi n} = 1 \text{ always})$$

$$\text{IDFT}[X^*(K)] = \frac{1}{N} \sum_{k=0}^{N-1} \left[X(k) e^{-j\left(\frac{2\pi k(N-n)}{N}\right)} \right]^* = x^*(N-n) \quad \text{hence} \quad \text{DFT}[x^*(N-n)] = [X^*(K)]$$

13. Show that the DFT coefficients $X(N+k)$ and $X(N-k)$ of N valued real sequence $x(n)$ are complex conjugates

Solution: $X[k] = \sum_{n=0}^{N-1} x(n) W_N^{kn}$

and $X[N+k] = \sum_{n=0}^{N-1} x(n) W_N^{(N+k)n} = \sum_{n=0}^{N-1} x(n) W_N^{kn} W_N^{Nn} = \sum_{n=0}^{N-1} x(n) W_N^{kn} (1) = X[k]$

Similarly $X[N-k] = \sum_{n=0}^{N-1} x(n) W_N^{(N-k)n} = \sum_{n=0}^{N-1} x(n) W_N^{-kn} W_N^{Nn} = \sum_{n=0}^{N-1} x(n) W_N^{-kn} (1) = X^*[k]$

Hence the DFT coefficients $X[N+k]$ and $X[N-k]$ are complex conjugates.

14. The 5 point DFT's of a complex sequence $x(n)$ are $\{j, (1+j), (1+j^2), (2+j^2), (4+j)\}$.
Compute $Y(k)$ if $y(n) = x^*(n)$

Solution: Given $N = 5$; and $X(K) = \{j, (1+j), (1+j^2), (2+j^2), (4+j)\}$

Given $y(n) = x^*(n)$

hence $Y(k) = \text{DFT}[x^*(n)] = X^*(N-k)$ ----- from complex conjugate property

$$K = 0; \rightarrow Y(0) = X^*(5) = X^*(0) = [j]^* = -j$$

$$K = 1; \rightarrow Y(1) = X^*(5-1) = X^*(4) = [4+j]^* = 4-j$$

$$K = 2; \rightarrow Y(2) = X^*(5-2) = X^*(3) = [2+j^2]^* = 2-j^2$$

$$K = 3; \rightarrow Y(3) = X^*(5-3) = X^*(2) = [1+j^2]^* = 1-j^2$$

$$K = 4; \rightarrow Y(4) = X^*(5-4) = X^*(1) = [1+j]^* = 1-j$$

$$Y(k) = \{-j, (4-j), (2-j^2), (1-j^2), (1-j)\}$$

15. If $x(n) \xleftrightarrow{\text{DFT}} X(k)$ then find the IDFT of $X^*(N-k)$

Solution: $X[k] = \sum_{n=0}^{N-1} x(n) W_N^{kn}$ or $X[N-k] = \sum_{n=0}^{N-1} x(n) W_N^{(N-k)n} = \sum_{n=0}^{N-1} x(n) W_N^{Nn} W_N^{-kn}$

$$X(N-k) = \sum_{n=0}^{N-1} x(n) W_N^{-kn} \quad \text{Taking conjugate on both sides}$$

$$X^*(N-k) = \sum_{n=0}^{N-1} [x(n) W_N^{-kn}]^* = \sum_{n=0}^{N-1} x^*(n) W_N^{kn} = \text{DFT}[x^*(n)] \quad \text{or} \quad \text{IDFT}[X^*(N-k)] = x^*(n)$$

N - point DFT of two real sequences using single N – point DFT

Let $g(n)$ and $h(n)$ are the two real valued sequences with DFT's $G(k)$ and $H(k)$ respectively. Let $x(n)$ be a complex sequence such that $x(n) = g(n) + j h(n)$, then $X(k) = G(k) + j H(k)$

$$g(n) = \frac{x(n) + x^*(n)}{2} \quad \text{and} \quad h(n) = \frac{x(n) - x^*(n)}{2j} \quad G(k) = \frac{X(k) + X^*(N-k)}{2} \quad \text{and} \quad H(k) = \frac{X(k) - X^*(N-k)}{2j}$$

16. Find the 4 point DFT's of two sequences $g(n)$ and $h(n)$ defined below using a single 4 point DFT
 $g(n) = \{1, 2, 0, 1\}$; $h(n) = \{2, 2, 1, 1\}$ and $x(n) = g(n) + j h(n)$

Solution: $x(n) = g(n) + j h(n) = \{(1+j2), (2+j2), j1, (1+j1)\}$

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} = \sum_{n=0}^3 x(n) W_4^{kn}$$

$$X(k) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -j & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1+j2 \\ 2+j2 \\ j1 \\ 1+j1 \end{bmatrix} = \begin{bmatrix} 4+j6 \\ 2 \\ -2 \\ j2 \end{bmatrix}$$

$$X(k) = \{(4+j6), 2, -2, j2\}$$

$$X^*(N-k) = X^*(4-k) = \{(4-j6), -j2, -2, 2\}$$

$$G(k) = \frac{X(k) + X^*(N-k)}{2} = \left(\frac{\{(4+j6), 2, -2, j2\} + \{(4-j6), -j2, -2, 2\}}{2} \right) = \{4, (1-j), -2, (1+j)\}$$

$$H(k) = \frac{X(k) - X^*(N-k)}{2j} = \left(\frac{\{(4+j6), 2, -2, j2\} - \{(4-j6), -j2, -2, 2\}}{2j} \right) = \{6, (1-j), 0, (1+j)\}$$

Circular Correlation Property

If $x(n) \xleftarrow{\text{DFT}} X(k)$ and $y(n) \xleftarrow{\text{DFT}} Y(k)$, then

$$\tilde{r}_{xy}(m) \xleftarrow{\text{DFT}} \tilde{R}_{xy}(k) = X(k) Y^*(k)$$

$$\tilde{r}_{xy}(m) = \sum_{n=0}^{N-1} x(n) y^*((n-m))_N$$

where $\tilde{r}_{xy}(m)$ is the circular cross correlation between $x(n)$ and $y(n)$

Proof:

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\left(\frac{2\pi kn}{N}\right)} \quad Y(k) = \sum_{p=0}^{N-1} y(p) e^{-j\left(\frac{2\pi kp}{N}\right)}$$

$$\text{IDFT}[X(k)Y^*(k)] = \frac{1}{N} \sum_{k=0}^{N-1} [X(k)Y^*(k)] e^{\frac{j2\pi kn}{N}} = \frac{1}{N} \sum_{k=0}^{N-1} [X(k)Y^*(k)] e^{\frac{j2\pi km}{N}} \quad \text{put } n = m$$

$$\text{IDFT}[X(k)Y^*(k)] = \frac{1}{N} \sum_{k=0}^{N-1} \left[\sum_{n=0}^{N-1} x(n) e^{-j\left(\frac{2\pi kn}{N}\right)} \right] \sum_{p=0}^{N-1} \left[y(p) e^{-j\left(\frac{2\pi kp}{N}\right)} \right]^* e^{\frac{j2\pi km}{N}}$$

$$\text{IDFT}[X(k)Y^*(k)] = \frac{1}{N} \sum_{n=0}^{N-1} x(n) \sum_{p=0}^{N-1} y^*(p) \sum_{k=0}^{N-1} e^{\frac{j2\pi k(m+p-n)}{N}}$$

$$\text{Put } (m+p-n) = qN$$

$$p = n - m + qN \text{ and } n = m + p - qN$$

$$\text{consider } \sum_{k=0}^{N-1} e^{\frac{j2\pi k(m+p-n)}{N}} = \sum_{k=0}^{N-1} e^{\frac{j2\pi kqN}{N}} = \sum_{k=0}^{N-1} 1^k = N$$

$$\text{Consider } \sum_{p=0}^{N-1} y^*(p) = \sum_{n=0}^{N-1} y^*(n - m + qN) = \sum_{n=0}^{N-1} y^*(n - m, \text{mod } N) = \sum_{n=0}^{N-1} y^*((n - m))_N$$

$$\text{IDFT}[X(k)Y^*(k)] = \frac{1}{N} \sum_{n=0}^{N-1} x(n) \sum_{n=0}^{N-1} y^*((n - m))_N N = \sum_{n=0}^{N-1} x(n) y^*((n - m))_N = \tilde{r}_{xy}(m)$$

$$X(k)Y^*(k) = \text{DFT} \left[\sum_{n=0}^{N-1} x(n) y^*((n - m))_N \right] = \text{DFT}[\tilde{r}_{xy}(m)] = \tilde{R}_{xy}(k)$$

$$\tilde{R}_{xy}(k) = X(k) \cdot Y^*(k)$$

Note: When $x(n) = y(n)$, then the correlation is called Circular Auto-correlation.

$$\tilde{r}_{xx}(m) \leftarrow \text{DFT} \rightarrow \tilde{R}_{xx}(k) = X(k) X^*(k) = |X(k)|^2$$

17. If $x_1(n) = \cos(2\pi n/N)$ and $x_2(n) = \sin(2\pi n/N)$ Determine the N point circular correlation of
(i) $x_1(n)$ and $x_2(n)$ (ii) $x_1(n)$ and $x_1(n)$ (iii) $x_2(n)$ and $x_2(n)$

$$\text{Solution: } x_1(n) = \cos\left(\frac{2\pi n}{N}\right) = \frac{e^{\frac{j2\pi n}{N}} + e^{-\frac{j2\pi n}{N}}}{2} = \frac{(W_N)^{-n} + (W_N)^n}{2}$$

$$X_1(k) = \sum_{n=0}^{N-1} \cos\left(\frac{2\pi n}{N}\right) (W_N)^{kn} = \sum_{n=0}^{N-1} \left(\frac{(W_N)^{-n(k-1)} + (W_N)^{n(k+1)}}{2} \right) = \frac{N}{2} [(\delta(k-1) + \delta(k+1))]$$

$$x_2(n) = \sin\left(\frac{2\pi n}{N}\right) = \frac{e^{\frac{j2\pi n}{N}} - e^{-\frac{j2\pi n}{N}}}{2j} = \frac{(W_N)^{-n} - (W_N)^n}{2}$$

$$X_2(k) = \sum_{n=0}^{N-1} \sin\left(\frac{2\pi n}{N}\right) (W_N)^{kn} = \sum_{n=0}^{N-1} \left(\frac{(W_N)^{-n(k-1)} - (W_N)^{n(k+1)}}{2j} \right) = \frac{N}{2j} [(\delta(k-1) - \delta(k+1))]$$

$$(i) X_1(k).X_2^*(k) = \tilde{R}_{X_1X_2}(k) = \frac{N}{2} [(\delta(k-1) + \delta(k+1)).(-\frac{N}{2j} [(\delta(k-1) - \delta(k+1))])$$

$$\tilde{R}_{X_1X_2}(k) = -\frac{N}{2} \left(\frac{N}{2j} [\delta(k-1) - \delta(k+1)] \right) \quad \text{Taking IDFT on both sides}$$

$$\tilde{r}_{x_1x_2}(m) = \text{IDFT} [X_1(k).X_2^*(k)] = -\frac{N}{2} \text{IDFT} \left(\frac{N}{2j} [\delta(k-1) - \delta(k+1)] \right) = -\frac{N}{2} \sin\left(\frac{2\pi n}{N}\right)$$

$$(ii) X_1(k).X_1^*(k) = \tilde{R}_{X_1X_1}(k) = \frac{N}{2} [(\delta(k-1) + \delta(k+1)).\left(\frac{N}{2} [(\delta(k-1) + \delta(k+1))]\right)$$

$$\tilde{R}_{X_1X_2}(k) = \frac{N}{2} \left(\frac{N}{2} [\delta(k-1) + \delta(k+1)] \right) \quad \text{Taking IDFT on both sides}$$

$$\tilde{r}_{x_1x_1}(m) = \text{IDFT} [X_1(k).X_1^*(k)] = \frac{N}{2} \text{IDFT} \left(\frac{N}{2} [\delta(k-1) + \delta(k+1)] \right) = \frac{N}{2} \cos\left(\frac{2\pi n}{N}\right)$$

$$(iii) X_2(k).X_2^*(k) = \tilde{R}_{X_2X_2}(k) = \frac{N}{2j} [(\delta(k-1) - \delta(k+1)).\left(-\frac{N}{2j} [(\delta(k-1) - \delta(k+1))]\right)$$

$$\tilde{R}_{X_2X_2}(k) = \frac{N}{2} \left(\frac{N}{2} [\delta(k-1) - \delta(k+1)] \right) \quad \text{Taking IDFT on both sides}$$

$$\tilde{r}_{x_2x_2}(m) = \text{IDFT} [X_2(k).X_2^*(k)] = \frac{jN}{2} \text{IDFT} \left(\frac{N}{2j} [\delta(k-1) - \delta(k+1)] \right) = j\frac{N}{2} \sin\left(\frac{2\pi n}{N}\right)$$

Parseval's Theorem

If $x(n) \xleftrightarrow{\text{DFT}_N} X(k)$ $y(n) \xleftrightarrow{\text{DFT}_N} Y(k)$ then

$$\sum_{n=0}^{N-1} x(n) y^*(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) Y^*(k)$$

Proof: Consider the RHS of the Parseval's relation

$$\text{RHS} = \frac{1}{N} \sum_{k=0}^{N-1} X(k) Y^*(k) = \frac{1}{N} \sum_{k=0}^{N-1} \left[\sum_{n=0}^{N-1} x(n) e^{-j\left(\frac{2\pi kn}{N}\right)} \right] Y^*(k) = \sum_{k=0}^{N-1} x(n) \left[\frac{1}{N} \sum_{n=0}^{N-1} Y^*(k) e^{-j\left(\frac{2\pi kn}{N}\right)} \right]$$

$$\frac{1}{N} \sum_{k=0}^{N-1} X(k) Y^*(k) = \sum_{k=0}^{N-1} x(n) \left[\frac{1}{N} \sum_{n=0}^{N-1} Y(k) e^{j\left(\frac{2\pi kn}{N}\right)} \right]^* = \sum_{n=0}^{N-1} x(n) y^*(n) = \text{LHS}$$

Note: when $x(n) = y(n)$ then $\sum_{n=0}^{N-1} |x(n)|^2 = E = \frac{1}{N} \sum_{k=0}^{N-1} |X(k)|^2$ — — — gives the energy of the sequence

18. Let $x(n) = \{ 2, 1, 1, 0, 3, 2, 0, 3, 4, 6 \}$ with 10 point DFT of $X[k]$, evaluate the following without explicitly computing DFT

$$(i) X[0] \quad (ii) X[5] \quad (iii) \sum_{k=0}^9 |X(k)|^2 \quad (iv) \sum_{k=0}^9 |X(k)| \quad (v) \sum_{k=0}^9 |X(k)| e^{j\left(\frac{4\pi k}{5}\right)}$$

Solution: $X[k] = x(0) + x(1)W_N^k + x(2)W_N^{2k} + \dots + x(9)W_N^{9k}$

(i) $X[0] = 2 + 1 + 1 + 0 + 3 + 2 + 0 + 3 + 4 + 6 = 22$

(ii) $X[5] = 2 + 1W_{10}^5 + 1W_{10}^{10} + 3W_{10}^{20} + 2W_{10}^{25} + 3W_{10}^{35} + 4W_{10}^{40} + 6W_{10}^{45}$

$X[5] = 2 - 1 + 1 + 3 - 2 - 3 + 4 - 6 = -2$

(iii) $\sum_{n=0}^{N-1} |X(k)|^2 = N \sum_{n=0}^{N-1} |x(n)|^2$ from Parseval's theorem

$\sum_{n=0}^9 |X(k)|^2 = 10 \sum_{n=0}^9 |x(n)|^2 = 10[2^2 + 1^2 + 1^2 + 3^2 + 2^2 + 3^2 + 4^2 + 6^2] = 800 \text{ Joules}$

(iv) $x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X[k] [W_N]^{-kn}$ or $N x(n) = \sum_{k=0}^{N-1} X[k] [W_N]^{-kn}$

put $n = 0$ then; $\sum_{k=0}^9 X[k] = N x(0)$ or $\sum_{k=0}^9 X[k] = 10(2) = 20$

(v) $W_{10}^{4k} = e^{-j2\pi 4k/10} = e^{-j2\pi 2k/5} = W_5^{2k}$

$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X[k] [W_N]^{-kn}$ or $N x(n) = \sum_{k=0}^{N-1} X[k] [W_N]^{-kn}$

put $n = -4$ then; $10 x(-4) = \sum_{k=0}^9 X[k] [W_{10}]^{4k}$ or $\sum_{k=0}^9 X[k] e^{-j2\pi 2k/5} = 10x(N-4) = 10x(10-4) = 0$

20. For $x(n) = \{ 1, -2, 3, -4, 5, -6 \}$, without computing DFT, find the following

(i) $X[0]$ (ii) $X[3]$ (iii) $\sum_{n=0}^5 X[k]$ (iv) $\sum_{n=0}^5 |X[k]|^2$ (v) $\sum_{n=0}^5 (-1)^k X[k]$

Solution: $X[k] = x(0) + x(1)W_N^k + x(2)W_N^{2k} + \dots + x(5)W_N^{5k}$

(i) $X[0] = 1 - 2 + 3 - 4 + 5 - 6 = -3$

(ii) $X[3] = 1 - 2W_6^3 + 3W_6^6 - 4W_6^9 + 5W_6^{12} - 6W_6^{15}$

$X[3] = 1 + 2 + 3 + 4 + 5 + 6 = 21$

(iii) $X[k] = \sum_{n=0}^{N-1} x(n) [W_N]^{kn}$ put $n = 0$ then; $X[k] = \sum_{n=0}^{N-1} x(0) (1) = N x(0) = 6(1) = 6$

(iv) $\sum_{k=0}^{N-1} |X(k)|^2 = N \sum_{n=0}^{N-1} |x(n)|^2$ from Parseval's theorem

$\sum_{k=0}^5 |X(k)|^2 = 6 \sum_{n=0}^5 |x(n)|^2 = 6[1^2 + (-2)^2 + 3^2 + (-4)^2 + 5^2 + (-6)^2] = 582 \text{ Joules}$

(v) $(-1)^k = e^{j\pi k} = \left[e^{-j\frac{\pi}{3}} \right]^{-3k} = \left[e^{-j\frac{2\pi}{6}} \right]^{-3k} = W_6^{-3k}$

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X[k] [W_N]^{-kn} \quad \text{with } n = 3 \text{ and } N = 6 \text{ then} \quad x(3) = \frac{1}{6} \sum_{k=0}^5 X[k] [W_6]^{-3k}$$

$$\sum_{k=0}^5 X[k] (-1)^k = 6 x(3) = 6(-4) = -24$$

21. The first 8 samples of a 14 length real sequence $x(n)$ are

$\{12, (-1+3j), (3+4j), (1-5j), (-2+2j), (6+3j), (-2-3j), 10\}$ Find the remaining samples and evaluate

(i) $x(0)$ (ii) $x(7)$ (iii) $\sum_{k=0}^{13} x(n)$ (iv) $\sum_{k=0}^{13} |x(n)|^2$

Solution: Using the complex conjugate property $X(N-k) = X^*(k)$ here $N = 14$

$$X(14-k) = X^*(k)$$

$$k = 6; \quad X(8) = X^*(6) = (-2-3j)^* = (-2+3j),$$

$$k = 5; \quad X(9) = X^*(5) = (6+3j)^* = (6-3j)$$

$$k = 4; \quad X(10) = X^*(4) = (-2+2j)^* = (-2-2j)$$

$$k = 3; \quad X(11) = X^*(3) = (1-5j)^* = (1+5j),$$

$$k = 2; \quad X(12) = X^*(2) = (3+4j)^* = (3-4j)$$

$$k = 1; \quad X(13) = X^*(1) = (-1+3j)^* = (-1-3j)$$

$$(i) \quad x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X[k] [W_N]^{-kn} = \frac{1}{14} \sum_{k=0}^{13} X[k] [W_{14}]^{-kn}$$

$$\text{Put } n = 0; \rightarrow x(0) = \frac{1}{14} \sum_{k=0}^{13} X[k] = 2.286$$

$$(ii) \quad \text{Put } n = \frac{N}{2} = 7; \quad x(7) = \frac{1}{14} \sum_{k=0}^{13} X[k] e^{-j2\pi k 7/14} = \frac{1}{14} \sum_{k=0}^{13} X[k] (-1)^k = -0.857$$

$$(iii) \quad X(k) = \sum_{n=0}^{N-1} x[n] [W_N]^{kn} = \sum_{n=0}^{13} x[n] [W_{14}]^{kn}$$

$$\text{put } k = 0; \quad X(0) = \sum_{n=0}^{13} x[n] [W_{14}]^0 \quad \text{or} \quad \sum_{n=0}^{13} x[n] = X(0) = 12$$

$$(iv) \quad \sum_{n=0}^{N-1} |x(n)|^2 = \frac{1}{14} \sum_{k=0}^{13} |X(k)|^2 = 35.57 \text{ Joules}$$

22. Compute the energy of the N point sequence $x(n) = \cos(2\pi k_0 n/N)$ --- $0 \leq n \leq (N-1)$

Solution: $X[k] = \left(\frac{N}{2}\right) [\delta(k - k_0) + \delta(k + k_0)]$

$$\text{Energy } E = \frac{1}{N} \sum_{k=0}^{N-1} |X(k)|^2 = \frac{1}{N} \left(\left[\frac{N}{2}\right]^2 + \left[\frac{N}{2}\right]^2 \right) = \frac{N}{2} \text{ Joule}$$

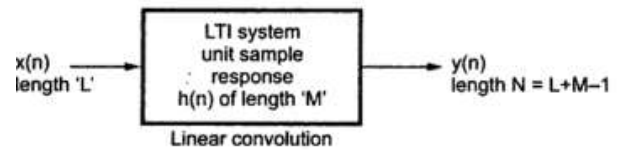
Applications of DFT: Linear Filtering Using DFT

Consider a finite duration sequence $x(n)$ of length L which excites an FIR filter of length M

$$x(n) = 0; \dots\dots n < 0 \text{ and } n \geq L$$

$$h(n) = 0; \dots\dots n < 0 \text{ and } n \geq M$$

The output sequence $y(n)$ of the FIR filter can be expressed in time domain as the linear convolution of $x(n)$ and $h(n)$ which has $(L + M - 1)$ samples.



$$y(n) = x(n) * h(n) = \sum_{k=0}^{N-1} h(k)x(n-k) \quad \text{----- (A)}$$

The frequency domain equivalent of equation (A) is $Y(\Omega) = X(\Omega)H(\Omega)$ ----- (B)

The output sequence $y(n)$ can be determined from its spectrum via inverse FT. As the spectrum is continuous, it cannot be processed on digital computer. On the other hand, DFT can be used to overcome the above problem and it allows linear filtering in frequency domain.

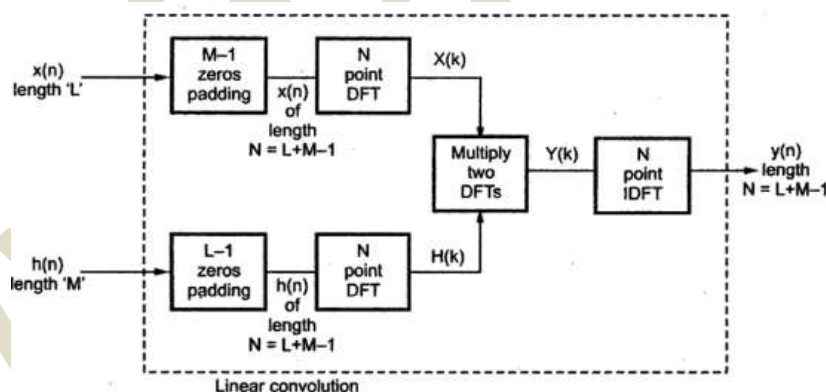
The DFT of size $N \geq (L + M - 1)$ is required to represent $y(n)$ uniquely in frequency domain.

$$Y(k) = Y(\Omega)_{\Omega=\frac{2\pi k}{N}} = [X(\Omega)H(\Omega)]_{\Omega=\frac{2\pi k}{N}} = X(k)H(k) \quad \text{where } k = 0, 1, 2, \dots, N-1$$

The sequence $y(n)$ can be obtained by taking the IDFT of $Y[k]$

$$y(n) = \text{IDFT}[Y(k)] = \text{IDFT}[X(k)H(k)] \quad ; \quad k = 0, 1, 2, \dots, (N-1)$$

Hence Linear convolution can be obtained from DFT and the above operation is implemented as shown below



Linear Convolution using Circular Convolution:

Consider the equation $Y(k) = X(k)H(k)$ where $k = 0, 1, 2, \dots, N-1$ ----- (a)

From the circular convolution property of DFT,

$$x_1(n) \otimes_N x_2(n) = X_1(k)X_2(k) \quad \text{----- (b)}$$

From equation (b), it can be concluded that multiplication of two DFT's in frequency domain is equivalent to circular convolution in time domain.

Applying this property to equation (a) results

$$y(n) = x(n) \otimes_N h(n)$$

Thus N -point circular convolution of $x(n)$ and $h(n)$ gives $y(n)$.

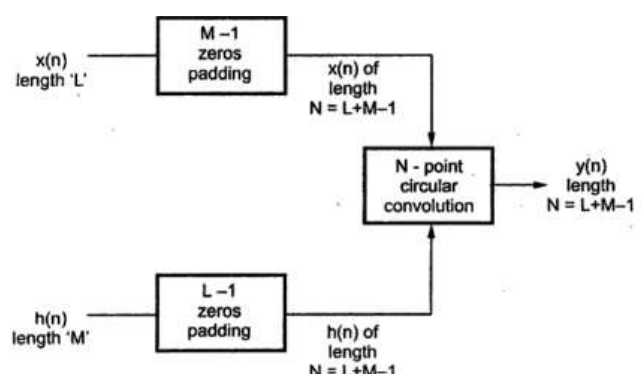


Fig. Linear convolution implemented by circular convolution

Linear convolution can be obtained from circular convolution if $N = (L + M - 1)$

1. A FIR filter has an impulse response of $h(n) = \{1, 2, 3\}$. Determine the response of the filter to the input sequence $x(n) = \{1, 2\}$. Use DFT and IDFT. Verify the result using direct computation of linear convolution.

Solution: $x(n) = \{1, 2\}$ $M = 2$

$h(n) = \{1, 2, 3\} = \{1, 2, 3\}$ $L = 3$

Make the length of the sequences as $N = M + L - 1 = 2 + 3 - 1 = 4$

$x(n) = \{1, 2, 0, 0\}$ and $h(n) = \{1, 2, 3, 0\}$

$X(k) = \{3, (1 - 2j), -1, (1 + 2j)\}$

$H(k) = \{6, (-2 - 2j), 2, (-2 + 2j)\}$

$$X(k) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 1-2j \\ -1 \\ 1+2j \end{bmatrix}$$

$$H(k) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ -2-2j \\ 2 \\ -2+2j \end{bmatrix}$$

$Y(k) = X(k) \cdot H(k) = \{18, -6+2j, -2, -6-2j\}$

$$y(n) = \frac{1}{N} \sum_{k=0}^{N-1} Y(k) W_N^{*} = \frac{1}{4} \sum_{k=0}^3 Y(k) W_N^{*} = (1/4) \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 18 \\ -6+2j \\ -2 \\ -6-2j \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 7 \\ 6 \end{bmatrix}$$

Hence $y(n) = \{1, 4, 7, 6\}$

Verification: Using linear convolution

$$\begin{array}{c|ccc} \xrightarrow{h(n)} & 1 & 2 & 3 \\ \xrightarrow{x(n)} \downarrow 1 & 1 & 2 & 3 \\ & 2 & 4 & 6 \end{array} \quad y(n) = \{1, 4, 7, 6\}$$

2. Determine the response of the FIR filter whose unit sample response is given as $h(n) = \{1, 2\}$ when the input applied is $x(n) = \{2, 1\}$. Use circular convolution and verify your result using linear convolution.

Solution: (i) Using Circular convolution $M = L = 2$; hence $N = M + L - 1 = 2 + 2 - 1 = 3$

$x(n) = \{2, 1, 0\}$ and $h(n) = \{1, 2, 0\}$

$y(n) = \{2, 5, 2\}$

Index	$x(m)$	$h(m-n)$	$y(n)$
0	$\{2, 1, 0\}$	$\{1, 0, 2\}$	$\{2 + 0 + 0\} = 2$
1	$\{2, 1, 0\}$	$\{2, 1, 0\}$	$\{4 + 1 + 0\} = 5$
2	$\{2, 1, 0\}$	$\{0, 2, 1\}$	$\{0 + 2 + 0\} = 2$

(ii) Verification using linear convolution

$y(n) = \{2, 5, 2\}$

$$\begin{array}{c|cc} \xrightarrow{x(n)} & 2 & 1 \\ \xrightarrow{h(n)} \downarrow 1 & 2 & 1 \\ & 4 & 2 \end{array}$$

3. Determine the output $y(n)$ for a system with $h(n) = \{1, 2\}$ to an input $x(n) = \{1, 2, 0, 4\}$ using circular convolution.

Solution: $x(n) = \{1, 2, 0, 4\}$ $M = 4$; $h(n) = \{1, 2\}$ $L = 2$

Modify the length of the sequences $N = M + L - 1 = 4 + 2 - 1 = 5$

$x(n) = \{1, 2, 0, 4, 0\}$ and $h(n) = \{1, 2, 0, 0, 0\}$

$$y(n) = x(n) \otimes h(n) = \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \\ 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 4 \\ 4 \\ 8 \end{bmatrix}$$

$$y(n) = \{1, 4, 4, 4, 8\}$$

4. Determine the output of the system if the input $x(n) = \delta(n) + 3\delta(n-1) + 2\delta(n-2) - \delta(n-3) + \delta(n-4)$. The discrete LTI system has an impulse response $h(n) = 2\delta(n) - \delta(n-1)$.

Solution: Given $x(n) = \{1, 3, 2, -1, 1\}$ $M = 4$; and $h(n) = \{2, -1\}$ $L = 2$

Modify the length of the sequences $N = M + L - 1 = 5 + 2 - 1 = 6$

$x(n) = \{1, 3, -1, 1, 0\}$ and $h(n) = \{2, -1, 0, 0, 0, 0\}$

$$y(n) = x(n) \otimes h(n) = \begin{bmatrix} 1 & 0 & 1 & -1 & 2 & 3 \\ 3 & 1 & 0 & 1 & -1 & 2 \\ 2 & 3 & 1 & 0 & 1 & -1 \\ -1 & 2 & 2 & 1 & 0 & 1 \\ 1 & -1 & 2 & 3 & 1 & 0 \\ 0 & 1 & -1 & 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 1 \\ -4 \\ 3 \\ -1 \end{bmatrix}$$

$$y(n) = \{2, 5, 1, -4, 3, -1\}$$

5. Given $x(n) = \{1, 2, 3, 4\}$ and $h(n) = \{1, 2, 2\}$ Compute (i) Circular convolution (ii) Linear convolution (iii) Linear convolution using circular convolution.

6. The impulse response of the LTI system is $\{1, -2, 1\}$. Determine the output response of the system if the input to the system is $\{1, 2, 1\}$. Use circular convolution.

Linear Filtering of Long Duration Sequences using DFT (Sectional Convolution)

Linear filtering can be implemented using DFT. Sometimes if the input data sequence is $x(n)$ is very in long duration (like speech, ECG signal etc), the long duration sequence is segmented into short duration blocks and the successive input sequence blocks are filtered one at a time using DFT. Since convolution (filtering) is linear, the corresponding output blocks are fitted together to generate combined output sequence.

Advantages of using DFT for long duration sequences:

DFT provides more simpler approach compared to complicated calculations involved in linear convolution.

Linear filtering using DFT is computationally efficient because of FFT algorithms

There are two methods are available based on this approach – Overlap Save and Overlap Add methods

Overlap-Save Method:

Let the length of the input sequence is L_s and the length of the impulse response is M .

(1) The input sequence is divided into blocks of data size $N = L + M - 1$.

(2) Each block consists of last $(M - 1)$ data points of previous block followed by L new data points form a new data sequence of length $N = L + M - 1$.

(3) For the first block of data, the first $(M - 1)$ points are set to zero. Thus the blocks of data sequence are

$$x_1(n) = \{ \underbrace{0, 0, \dots, 0}_{M-1 \text{ zeros}}, x(0), x(1), x(2), \dots, x(L-1) \}$$

$$x_2(n) = \{ \underbrace{x(L-M+1), \dots, x(L-1)}_{M-1 \text{ data points from } x_1(n)}, \underbrace{x(L), \dots, x(2L-1)}_{L \text{ new data points}} \}$$

$$x_3(n) = \{ x(2L-M+1), \dots, x(2L-1), x(2L), \dots, x(3L-1) \} \text{ and so on.}$$

(4) Now the impulse response of the FIR filter is increased in length to N by appending $(L - 1)$ zeros

(5) N -point circular convolution of each input block with $h(n)$ is computed.

$$y_1(n) = x_1(n) *_{\text{N}} h(n); \quad y_2(n) = x_2(n) *_{\text{N}} h(n); \quad y_3(n) = x_3(n) *_{\text{N}} h(n) \text{ and so on.}$$

(6). Discard the first $M - 1$ data points in every filtered output and the remaining data points from successive sections are abutted to construct the filtered output.

Illustration: Let $x(n)$ is of length $L_s = 15$ and $h(n)$ be of length $M = 3$. The Length of the each block of input sequence is 5.

The input sequences are $x_1(n) = \{ 0, 0, x(0), x(1), x(2) \}$

$x_2(n) = \{ x(1), x(2), x(3), x(4), x(5) \}$ and $x_6(n) = \{ x(13), x(14), 0, 0, 0 \}$

$h(n) = \{ h(0), h(1), h(2), 0, 0 \}$

The 5 point circular convolution of $x_i(n)$ and $h(n)$ gives $y_i(n)$ in which the first $(M - 1)$ points are discarded.

$$y_1(n) = x_1(n) \otimes_{\text{N}} h(n) = \{ y_1(0), y_1(1), y_1(2), y_1(3), y_1(4) \}$$

$$y_2(n) = x_2(n) \otimes_{\text{N}} h(n) = \{ y_2(0), y_2(1), y_2(2), y_2(3), y_2(4) \}$$

$$y_6(n) = x_6(n) \otimes_{\text{N}} h(n) = \{ y_6(0), y_6(1), y_6(2), y_6(3), y_6(4) \}$$

$$\text{Hence } y(n) = \{ y_1(2), y_1(3), y_1(4), y_2(2), y_2(3), y_2(4), \dots, y_6(2), y_6(3), y_6(4) \}$$

Overlap-Add Method

Let the length of the input sequence is L_s and the length of the impulse response is M .

(1) The input sequence is divided into blocks of data size L and $M - 1$ zeros are appended to it to make the data size of $N = L + M - 1$.

The data blocks are represented as

$$x_1(n) = \{ x(0), x(1), \dots, x(L-1), \underbrace{0, 0, \dots, 0}_{(M-1) \text{ zeros}} \}$$

$$x_2(n) = \{ x(L), x(L+1), \dots, x(2L-1), \underbrace{0, 0, \dots, 0}_{(M-1) \text{ zeros}} \}$$

$$x_3(n) = \{ x(2L), x(2L+1), \dots, x(3L-1), \underbrace{0, 0, \dots, 0}_{(M-1) \text{ zeros}} \}$$

(2) Now the impulse response of the FIR filter is increased in length to N by appending $(L - 1)$ zeros.

(3) Perform sectional N-point circular convolution of each input block with $h(n)$

$$y_1(n) = x_1(n) *_N h(n); \quad y_2(n) = x_2(n) *_N h(n); \quad y_3(n) = x_3(n) *_N h(n) \quad \text{and so on.}$$

(4) Since each block is terminated with $(M - 1)$ zeros, the last $(M - 1)$ samples from each filtered output must be overlapped and added to first $(M - 1)$ samples of successive filtered output to get the final filtered output. Hence the name overlap add.

The output blocks are of the form

$$y_1(n) = \{ y_1(0), y_1(1), \dots, y_1(L - 1), y_1(L), 0, \dots, y_1(N - 1) \}$$

$$y_2(n) = \{ y_2(0), y_2(1), \dots, y_2(L - 1), y_2(L), 0, \dots, y_2(N - 1) \}$$

$$y_3(n) = \{ y_3(0), y_3(1), \dots, y_3(L - 1), y_3(L), 0, \dots, y_3(N - 1) \} \dots\dots$$

$$\text{The output sequence } y(n) = \{ y_1(0), y_1(1), \dots, y_1(N - 1) + y_2(M - 1), y_2(M), \dots, y_3(N - 1) \}$$

1. Find the output $y(n)$ of a filter whose impulse response is $h(n) = \{ 1, 1, 1 \}$ and input signal $x(n) = \{ 3, -1, 0, 1, 3, 2, 0, 1, 2, 1 \}$ using (i) Overlap-Save method (ii) Overlap – Add method.

Solution (i) Overlap – Save Method

Given $h(n) = \{ 1, 1, 1 \}$ $M = 3$, hence $M - 1 = 2$

Let the block size is of length $N = 5$ so that $L = N - M + 1 = 5 - 3 + 1 = 3$

Modify $h(n)$ such that its length becomes N by padding $(L - 1)$ zeros ; $h(n) = \{ 1, 1, 1, 0, 0 \}$

Now make $x(n)$ into segments of length 5 as shown

$$x_1(n) = \{ 0, 0, 3, -1, 0 \}; \quad x_2(n) = \{ \underline{-1}, \underline{0}, 1, 3, 2 \}; \quad x_3(n) = \{ \underline{3}, \underline{2}, 0, 1, 2 \}; \quad x_4(n) = \{ \underline{1}, \underline{2}, 1, 0, 0 \}$$

$$y_1(n) = x_1(n) \otimes h(n) = \{ -1, 0, 3, 2, 2 \}$$

$$\begin{vmatrix} 0 & 0 & -1 & 3 & 0 \\ 0 & 0 & 0 & -1 & 3 \\ 3 & 0 & 0 & 0 & -1 \\ -1 & 3 & 0 & 0 & 0 \\ 0 & -1 & 3 & 0 & 0 \end{vmatrix} \begin{vmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \end{vmatrix} = \begin{vmatrix} 0+0-1+0+0 \\ 0+0+0+0+0 \\ 3+0+0+0+0 \\ -1+3+0+0+0 \\ 0-1+3+0+0 \end{vmatrix} = \begin{vmatrix} -1 \\ 0 \\ 3 \\ 2 \\ 2 \end{vmatrix}$$

$$\text{Similarly } y_2(n) = \{ \mathbf{4}, \mathbf{1}, 0, 4, 6 \} \quad y_3(n) = \{ \mathbf{6}, \mathbf{7}, 5, 3, 3 \} \quad y_4(n) = \{ \mathbf{1}, \mathbf{3}, 4, 3, 1 \}$$

Discard the first $(M - 1) = 2$ samples from each convolution output and build the final filter output from the remaining samples
(highlighted samples are discarded from the output)

$$\text{Hence } y(n) = \{ 3, 2, 2, 0, 4, 6, 5, 3, 3, 4, 3, 1 \}$$

(ii) Overlap – Add Method

Given $h(n) = \{ 1, 1, 1 \}$ $M = 3$, hence $M - 1 = 2$

Let the block size is of length $N = 5$ so that $L = N - M + 1 = 5 - 3 + 1 = 3$

Modify $h(n)$ such that its length becomes N by padding $(L - 1)$ zeros ; $h(n) = \{ 1, 1, 1, 0, 0 \}$

Now make $x(n)$ into segments of length 5 by padding $(M - 1) = 2$ zeros as shown

$$x(n) = \{ 3, -1, 0, 1, 3, 2, 0, 1, 2, 1 \}$$

$$x_1(n) = \{ 3, -1, 0, 0, 0 \}; \quad x_2(n) = \{ 1, 3, 2, 0, 0 \}; \quad x_3(n) = \{ 0, 1, 2, 0, 0 \}; \quad x_4(n) = \{ 1, 0, 0, 0, 0 \}$$

$$y_1(n) = h(n) \otimes x_1(n) = \begin{vmatrix} 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{vmatrix} \begin{vmatrix} 3 \\ -1 \\ 0 \\ 0 \\ 0 \end{vmatrix} = \begin{vmatrix} 3+0+0+0+0 \\ 3-1+0+0+0 \\ 3-1+0+0+0 \\ 0-1+0+0+0 \\ 0+0+0+0+0 \end{vmatrix} = \begin{vmatrix} 3 \\ 2 \\ 2 \\ -1 \\ 0 \end{vmatrix}$$

$$\text{Similarly } y_2(n) = \{ 1, 4, 6, 5, 2 \} \quad y_3(n) = \{ 0, 1, 3, 3, 2 \} \quad y_4(n) = \{ 1, 1, 1, 0, 0 \}$$

Overlap and add the last $(M - 1)$ and first $(M - 1)$ samples of each output to get the final filter output.

	3	2	2	-1	0									
				1	4	6	5	2						
							0	1	3	3	2			
										1	1	1	0	0
$y(n)$	3	2	2	0	4	6	5	3	3	4	3	1	0	0

$$\text{Hence } y(n) = \{ 3, 2, 2, 0, 4, 6, 5, 3, 3, 4, 3, 1 \}$$

2. Find the output $y(n)$ of a filter whose impulse response $h(n) = \{ 1, 2, 3, 4 \}$ and input signal to the filter is $x(n) = \{ 1, 2, 1, -1, 3, 0, 5, 6, 2, -2, -5, -6, 7, 1, 2, 0, 1 \}$ using overlap – add method with 6 point circular convolution

Solution: Given $N = 6$; $M = 4 \rightarrow L = N - M + 1 = 6 - 4 + 1 = 3$; $(M - 1) = 3$ zeros are padded each sequence.

$$\begin{aligned} x_1(n) &= \{ 1, 2, 1, 0, 0, 0 \} & x_2(n) &= \{ -1, 3, 0, 0, 0, 0 \} & x_3(n) &= \{ 5, 6, 2, 0, 0, 0 \} \\ x_4(n) &= \{ -2, -5, -6, 0, 0, 0 \} & x_5(n) &= \{ 7, 1, 2, 0, 0, 0 \} & \text{and } x_6(n) &= \{ 0, 1, 0, 0, 0, 0 \} \quad \text{and } h(n) = \{ 1, 2, 3, 4, 0, 0 \} \end{aligned}$$

$x_1(m)$						$h(n - m)$						$y_1(n)$					
1	2	1	0	0	0	1	0	0	4	3	2	$(1 + 0 + 0 + 0 + 0 + 0) = 1$					
1	2	1	0	0	0	2	1	0	0	4	3	$(2 + 2 + 1 + 0 + 0 + 0) = 4$					
1	2	1	0	0	0	3	2	1	0	0	4	$(3 + 4 + 1 + 0 + 0 + 0) = 8$					
1	2	1	0	0	0	4	3	2	1	0	0	$(4 + 6 + 2 + 0 + 0 + 0) = 12$					
1	2	1	0	0	0	0	4	3	2	1	0	$(0 + 8 + 3 + 0 + 0 + 0) = 11$					
1	2	1	0	0	0	0	0	4	3	2	1	$(0 + 0 + 4 + 0 + 0 + 0) = 4$					

$x_2(m)$						$h(n - m)$						$y_2(n)$					
-1	3	0	0	0	0	1	0	0	4	3	2	$(-1 + 0 + 0 + 0 + 0 + 0) = -1$					
-1	3	0	0	0	0	2	1	0	0	4	3	$(-2 + 3 + 0 + 0 + 0 + 0) = 1$					
-1	3	0	0	0	0	3	2	1	0	0	4	$(-3 + 6 + 0 + 0 + 0 + 0) = 3$					
-1	3	0	0	0	0	4	3	2	1	0	0	$(-4 + 9 + 0 + 0 + 0 + 0) = 5$					
-1	3	0	0	0	0	0	4	3	2	1	0	$(0 + 12 + 0 + 0 + 0 + 0) = 12$					
-1	3	0	0	0	0	0	0	4	3	2	1	$(0 + 0 + 4 + 0 + 0 + 0) = 4$					

$x_3(m)$						$h(n-m)$						$y_3(n)$					
5	6	2	0	0	0	1	0	0	4	3	2	$(5+0+0+0+0+0)=5$					
5	6	2	0	0	0	2	1	0	0	4	3	$(10+6+1+0+0+0)=16$					
5	6	2	0	0	0	3	2	1	0	0	4	$(15+12+2+0+0+0)=29$					
5	6	2	0	0	0	4	3	2	1	0	0	$(20+18+4+0+0+0)=42$					
5	6	2	0	0	0	0	4	3	2	1	0	$(0+24+6+0+0+0)=30$					
5	6	2	0	0	0	0	0	4	3	2	1	$(0+0+8+0+0+0)=8$					

$x_4(m)$						$h(n-m)$						$y_4(n)$					
-2	-5	-6	0	0	0	1	0	0	4	3	2	$(-2+0+0+0+0+0)=-2$					
-2	-5	-6	0	0	0	2	1	0	0	4	3	$(-4-5+0+0+0+0)=-9$					
-2	-5	-6	0	0	0	3	2	1	0	0	4	$(-6-10-6+0+0+0)=-22$					
-2	-5	-6	0	0	0	4	3	2	1	0	0	$(-8-15-12+0+0+0)=-35$					
-2	-5	-6	0	0	0	0	4	3	2	1	0	$(0-20-18+0+0+0)=-38$					
-2	-5	-6	0	0	0	0	0	4	3	2	1	$(0+0-24+0+0+0)=-24$					

$x_5(m)$						$h(n-m)$						$y_5(n)$					
7	1	2	0	0	0	1	0	0	4	3	2	$(7+0+0+0+0+0)=7$					
7	1	2	0	0	0	2	1	0	0	4	3	$(14+1+0+0+0+0)=15$					
7	1	2	0	0	0	3	2	1	0	0	4	$(21+2+2+0+0+0)=25$					
7	1	2	0	0	0	4	3	2	1	0	0	$(28+3+4+0+0+0)=35$					
7	1	2	0	0	0	0	4	3	2	1	0	$(0+4+6+0+0+0)=10$					
7	1	2	0	0	0	0	0	4	3	2	1	$(0+0+8+0+0+0)=8$					

$x_6(m)$						$h(n-m)$						$y_6(n)$					
0	1	0	0	0	0	1	0	0	4	3	2	$(0+0+0+0+0+0)=0$					
0	1	0	0	0	0	2	1	0	0	4	3	$(0+1+0+0+0+0)=1$					
0	1	0	0	0	0	3	2	1	0	0	4	$(0+2+0+0+0+0)=2$					
0	1	0	0	0	0	4	3	2	1	0	0	$(0+3+0+0+0+0)=3$					
0	1	0	0	0	0	0	4	3	2	1	0	$(0+4+0+0+0+0)=4$					
0	1	0	0	0	0	0	0	4	3	2	1	$(0+0+0+0+0+0)=0$					

	1	4	8	12	11	4													
				-1	1	3	5	12	0										
							5	16	29	42	30	8							
										-2	-9	-22	-35	-38	-24				
													7	15	25	35	10	8	
																0	1	2	3
y(n)	1	4	8	11	12	7	10	28	29	39	21	-14	-28	-23	1	35	11	10	3